Derivation of Normal Equations for Linear Regression

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1 Linear Regression

In these notes I want to explain you how the normal equations for linear regression are derived. For doing this, I would be using the vector and matrix calculus. Please do remember that you can find these details on the internet on different websites like wikipedia. The idea is same but the symbols might be different. I am going to stay consistant with my course contents.

We have defined the following hypothesis function:

$$h_\theta(x) = \theta_0 x_0 + \theta_1 x_1 + \ldots + \theta_n x_n, \tag{1}$$

where each x is a n-dimensional vector or observation. Our goal is to minimising the cost function $J(\Theta)$ using least squares as follows:

$$J(\Theta) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x)^{(i)} - y^{(i)})^2, \tag{2}$$

where $x^{(i)}$ is an i^{th} observation from m number of samples and $y^{(i)}$ is the ground truth (class label) for that i^{th} observation. While doing these calculations, the rule of thumb is to check the dimensions of the resulting vectors. This gives you a good idea of whether you have done the calculus correctly.

So here I redefine the vectors in our problem just as a reminder. The regression coefficients in our problem are θ defined as

$$\begin{pmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_n \end{pmatrix} \in \mathbb{R}^{n+1} \tag{3}$$

Each m^{th} input $\boldsymbol{x}$ is a vector of $n+1$ dimensions where we add $x_0 = 1$ for convenience. Therefore our hypothesis function becomes:

$$h_\theta(x) = \boldsymbol{\theta}^T \boldsymbol{x}, \tag{4}$$

where this equation represents the dot product of the two vectors. We define the '*design matrix*' are refered to as $\boldsymbol{X}$ using the following notation:

$$\boldsymbol{X} = \begin{bmatrix} \cdots x^{(1)T} \cdots \\ \cdots x^{(2)T} \cdots \\ \vdots \\ \cdots x^{(m)T} \cdots \end{bmatrix} \tag{5}$$

as a matrix of m rows, in which each row is the i^{th} sample (the vector $x^{(i)}$). With this, we can rewrite the least-squares cost as following, replacing the explicit sum by matrix multiplication:

$$J(\boldsymbol{\theta}) = \frac{1}{2m}(\boldsymbol{X\theta} - \boldsymbol{y})^T(\boldsymbol{X\theta} - \boldsymbol{y}) \tag{6}$$

Using some matrix transpose identities, we can simplify this a bit. I will throw away the fraction part $\frac{1}{2m}$, since we're going to compare a derivative to zero anyway. So here are the equations:

$$J(\boldsymbol{\theta}) = ((\boldsymbol{X\theta})^T - \boldsymbol{y}^T)(\boldsymbol{X\theta} - \boldsymbol{y}) \tag{7}$$

$$J(\boldsymbol{\theta}) = (\boldsymbol{X\theta})^T\boldsymbol{X\theta} - (\boldsymbol{X\theta})^T\boldsymbol{y} - \boldsymbol{y}^T(\boldsymbol{X\theta}) + \boldsymbol{y}^T\boldsymbol{y} \tag{8}$$

Note that $\boldsymbol{X\theta}$ is a vector, and so is $\boldsymbol{y}$. So when we multiply one by another, it doesn't matter what the order is (as long as the dimensions work out). So we can further simplify:

$$J(\boldsymbol{\theta}) = \boldsymbol{\theta}^T\boldsymbol{X}^T\boldsymbol{X\theta} - 2(\boldsymbol{X\theta})^T\boldsymbol{y} + \boldsymbol{y}^T\boldsymbol{y} \tag{9}$$

This is the matrix representation of the cost function J that we wished to minimise. Remember using the system of linear equations we had taken the parital derivative of the cost fucntion and equated it equal to zero. Using this derivative we used the Gradient descent algorithm to find the values of θ that will minimise this cost function.

Now we have to take the partial derivative with respect to vectors which is lightly uncomfortable, but I will try to explain it.

Lets work at the derivatives in terms of vectors. Lets consider the following fucntion:

$$f(v) = a^Tv \tag{10}$$

We know how to take the parital derivative of the above function by expanding it. So lets define the partial derivatives first:

$$\frac{\partial}{\partial v}f = \begin{pmatrix} \frac{\partial f}{\partial v_1} \\ \frac{\partial f}{\partial v_2} \\ \vdots \\ \frac{\partial f}{\partial v_n} \end{pmatrix} \tag{11}$$

We know how to solve this by opening the vectors as follows:

$$
\begin{aligned}
f(v) &= a^T v & (12)\\
&= a_1 v_1 + a_2 v_2 + \ldots + a_n v_n & (13)
\end{aligned}
$$

Now computing the partial derivatives of each component we get following:

$$
\begin{aligned}
\frac{\partial f}{\partial v_1} &= a_1\\
\frac{\partial f}{\partial v_2} &= a_2\\
&\vdots\\
\frac{\partial f}{\partial v_n} &= a_n & (14)\\
& & (15)
\end{aligned}
$$

So in terms of vectors, we can write the above equation as:

$$
\frac{\partial}{\partial v} f = \boldsymbol{a}. \tag{16}
$$

Lets keep working through the partial derivatives with respect to vectors. Lets define

$$
P(\theta) = 2(\boldsymbol{X\theta})^T \boldsymbol{y}. \tag{17}
$$

Over here please verify the dimensions of the above equation if the satisfy. Now lets expand the above equations:

$$
P(\theta) = 2 \times \left[\begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{pmatrix} \begin{pmatrix} \theta_1 \\ \theta_2 \\ \cdots \\ \theta_n \end{pmatrix} \right]^T \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix} \tag{18}
$$

Expanind the equations:

$$
P(x) = 2 \times \left[\begin{pmatrix} x_{11}\theta_1 + & \cdots & +x_{1n}\theta_n \\ x_{21}\theta_1 + & \cdots & +x_{2n}\theta_n \\ \cdots & \cdots & \cdots \\ x_{m1}\theta_1 + & \cdots & +x_{mn}\theta_n \end{pmatrix} \right]^T \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix} \tag{19}
$$

and now multiplying the matrix with vector $\boldsymbol{y}$:

$$
\begin{aligned}
P(\theta) = \;& 2(x_{11}\theta_1 + \cdots + x_{1n}\theta_n)y_1 + \\
& (x_{21}\theta_1 + \cdots + x_{2n}\theta_n)y_2 + \\
& \cdots + \\
& (x_{m1}\theta_1 + \cdots + x_{mn}\theta_n)y_m & (20)
\end{aligned}
$$

We can write the above equations in a much compact form using summations signs

$$\begin{aligned} P(x) &= 2\sum_{r=1}^{m}(x_{r1}\theta_1 + \cdots + x_{rn}\theta_n)y_r & (21)\\ &= 2\sum_{r=1}^{m} y_r \sum_{c=1}^{n} x_{rc}\theta_c & (22)\end{aligned}$$

Now start taking the partial derivatives of above equation by θ_i:

$$\begin{aligned} \frac{\partial P}{\partial \theta_1} &= 2(x_{11}y_1 + \cdots + x_{m1}y_m)\\ \frac{\partial P}{\partial \theta_2} &= 2(x_{12}y_1 + \cdots + x_{m2}y_m)\\ &\vdots\\ \frac{\partial P}{\partial \theta_n} &= 2(x_{1n}y_1 + \cdots + x_{mn}y_m)\end{aligned} \tag{23}$$

So we can write the above equations using the matrix forms as follows:

$$\frac{\partial}{\partial \boldsymbol{\theta}} P = 2\boldsymbol{X}\boldsymbol{y} \tag{24}$$

$$\begin{aligned} \frac{\partial}{\partial \boldsymbol{\theta}} P &= \frac{\partial}{\partial \boldsymbol{\theta}}(2(\boldsymbol{X}\boldsymbol{\theta})^T\boldsymbol{y})\\ &= 2\boldsymbol{X}\boldsymbol{y} \end{aligned} \tag{25}$$

Take a moment to convince yourself this is true. It's just collecting the individual components of $\boldsymbol{X}$ into a matrix and the individual components of $\boldsymbol{y}$ into a vector. Since $\boldsymbol{X}$ is a m-by-n matrix and y is a m-by-1 column vector, the dimensions work out and the result is a n-by-1 column vector.

So we've just computed the second term of the vector derivative of J. Now lets go back to the full definition of J and see how to compute the derivative of its first term. Lets call it Q

$$Q(\boldsymbol{\theta}) = \boldsymbol{\theta}^T\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{\theta} \tag{26}$$

This will be slightly more complex but here we go ... plz note there is a transpose of $\boldsymbol{X}$.

$$Q(\boldsymbol{\theta}) = (\theta_1 \cdots \theta_n)\begin{pmatrix} x_{11} & x_{21} & \cdots & x_{m1}\\ x_{12} & x_{22} & \cdots & x_{m2}\\ & \cdots & & \\ x_{1n} & x_{2n} & \cdots & x_{mn}\end{pmatrix}\begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n}\\ x_{21} & x_{22} & \cdots & x_{2n}\\ & \cdots & & \\ x_{m1} & x_{m2} & \cdots & x_{mn}\end{pmatrix}\begin{pmatrix}\theta_1\\ \theta_2\\ \vdots\\ \theta_n\end{pmatrix} \tag{27}$$

The centre two matrices are simply $\boldsymbol{X}$-squared matrix which is n-by-n. The element in row r and column c of this square matrix is:

$$\sum_{i=1}^{m} x_{ir}x_{ic}, \tag{28}$$

where $\boldsymbol{X}$-squared is a symmetric matrix. Lets call each element of this squared matrix as X^2_{rc}. Now the above equation becomes:

$$Q(\boldsymbol{\theta}) = (\theta_1 \cdots \theta_n) \begin{pmatrix} X^2_{11}\theta_1 + \cdots + X^2_{1n}\theta_n \\ X^2_{21}\theta_1 + \cdots + X^2_{2n}\theta_n \\ \cdots \\ X^2_{n1}\theta_1 + \cdots + X^2_{nn}\theta_n \end{pmatrix} \tag{29}$$

Now multiplying by θ we get:

$$\begin{aligned} Q(\boldsymbol{\theta}) = \quad & \theta_1(X^2_{11}\theta_1 + \cdots + X^2_{1n}\theta_n) + \\ & \theta_2(X^2_{21}\theta_1 + \cdots + X^2_{2n}\theta_n) + \\ & \cdots + \\ & \theta_n(X^2_{n1}\theta_1 + \cdots + X^2_{nn}\theta_n) \end{aligned} \tag{30}$$

Lets start calculating the partial derivatives as follows:

$$\frac{\partial Q}{\partial \theta_1} = (2\theta_1 X^2_{11} + \theta_2 X^2_{12} + \cdots + \theta_n X^2_{1n}) + \theta_2 X^2_{21} + \cdots + \theta_n X^2_{n1} \tag{31}$$

Remember $\boldsymbol{X}$ is a symmetric matrix, so $X^2_{12} = X^2_{21}$. Hence,

$$\frac{\partial Q}{\partial \theta_1} = 2\theta_1 X^2_{11} + 2\theta_2 X^2_{12} + \cdots + 2\theta_n X^2_{1n} \tag{32}$$

The remaining partial derivatives will also be similar.

Now writing the partial derivatives in the vector forms, we get:

$$\frac{\partial Q}{\partial \boldsymbol{\theta}} = 2\boldsymbol{X}^2\boldsymbol{\theta} = 2\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{\theta} \tag{33}$$

Hence,

$$\frac{\partial}{\partial \boldsymbol{\theta}} \boldsymbol{\theta}^T\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{\theta} = 2\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{\theta}. \tag{34}$$

Now lets look back it the matrix form of the cost fucntion J given in (9) and insert the partial derivatives.

$$\frac{\partial J}{\partial \boldsymbol{\theta}} = \frac{\partial}{\partial \theta}(\boldsymbol{\theta}^T \boldsymbol{X}^T \boldsymbol{X}\boldsymbol{\theta} - 2(\boldsymbol{X}\boldsymbol{\theta})^T \boldsymbol{y} + \boldsymbol{y}^T \boldsymbol{y}) \tag{35}$$

$$= 2\boldsymbol{X}^T \boldsymbol{X}\boldsymbol{\theta} - 2\boldsymbol{X}^T \boldsymbol{y} = 0 \tag{36}$$

OR

$$2\boldsymbol{X}^T \boldsymbol{X}\boldsymbol{\theta} = 2\boldsymbol{X}^T \boldsymbol{y} \tag{37}$$

Now assuming that matrix $\boldsymbol{X}^T \boldsymbol{X}$ is invertible, we get

$$\boldsymbol{\theta} = (\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{X}^T \boldsymbol{y} \tag{38}$$

which is the normal equation that we use for calculating the $\boldsymbol{\theta}$ empirically using matrix notations.

2 Derivation Using the Matrix Derivatives

Now that you know how to calculate the partial derivatives of matrices, lets have a look at how to solve the derivative of normal equations for linear regression. Lets recap the cost function for the linear regression:

$$J(\Theta) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x)^{(i)} - y^{(i)})^2, \tag{39}$$

The vectorised form of this equation is given as follows:

$$J(\boldsymbol{\theta}) = \frac{1}{2m} (\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y})^T (\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}) \tag{40}$$

Throwing away the fraction part, $\frac{1}{2m}$, is the simplified version of the equation

Using some matrix transpose identities, we can simplify this a bit. I will throw away the fraction part $\frac{1}{2m}$, since we're going to compare a derivative to zero anyway. So here are the equations:

$$J(\boldsymbol{\theta}) = ((\boldsymbol{X}\boldsymbol{\theta})^T - \boldsymbol{y}^T)(\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}) \tag{41}$$

$$J(\boldsymbol{\theta}) = (\boldsymbol{X}\boldsymbol{\theta})^T \boldsymbol{X}\boldsymbol{\theta} - (\boldsymbol{X}\boldsymbol{\theta})^T \boldsymbol{y} - \boldsymbol{y}^T (\boldsymbol{X}\boldsymbol{\theta}) + \boldsymbol{y}^T \boldsymbol{y} \tag{42}$$

Rearranging the above matrices and vectors, we get:

$$J(\boldsymbol{\theta}) = \boldsymbol{\theta}^T \boldsymbol{X}^T \boldsymbol{X}\boldsymbol{\theta} - 2\boldsymbol{y}^T (\boldsymbol{X}\boldsymbol{\theta}) + \boldsymbol{y}^T \boldsymbol{y} \tag{43}$$

Now our goal is simply to find the values of $\boldsymbol{theta}$ that minimises the cost function $J(\boldsymbol{\theta})$. To do that we use the concepts from our FCS where to minimise the function, all we need to do is to take the derivative of the function that we want minimise and put it equal to zero. To do that we got to remember one thing that here we are dealing with vectors and matrices not variables. So lets have a look at some of the rules that we are going to use.

2.1 Matrices Derivatives

If we have to take a derivative of a function of $\boldsymbol{\theta}, f(\boldsymbol{\theta})$ with respect to (w.r.t) $\boldsymbol{\theta}$, then the partial derivative is comes out to be:

$$\frac{\partial}{\partial \theta}(A\theta) = A^T \tag{44}$$

Secondly,

$$\frac{\partial}{\partial \theta}(\theta^T A\theta) = 2A^T\theta \tag{45}$$

2.2 Calculating the Derivatives

Now lets apply these functions on the cost function. Therefore:

$$\begin{aligned} \frac{\partial J(\theta)}{\partial \theta} &= \frac{\partial}{\partial \theta}(\boldsymbol{\theta}^T \boldsymbol{X}^T \boldsymbol{X}\boldsymbol{\theta} - 2\boldsymbol{y}^T(\boldsymbol{X}\boldsymbol{\theta}) + \boldsymbol{y}^T\boldsymbol{y})\frac{1}{2m} && (46) \\ &= (2\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{\theta} - 2\boldsymbol{X}^T\boldsymbol{y} + 0)\frac{1}{2m} && (47) \end{aligned}$$

All that we need to do now is to put the first derivative of the function equal to zero and solve for θ.

$$\begin{aligned} (2\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{\theta} - 2\boldsymbol{X}^T\boldsymbol{y} + 0)\frac{1}{2m} &= 0, && (48) \\ 2\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{\theta} &= 2\boldsymbol{X}^T\boldsymbol{y}, && (49) \\ \boldsymbol{X}^T\boldsymbol{X}\boldsymbol{\theta} &= \boldsymbol{X}^T\boldsymbol{y}, && (50) \\ \boldsymbol{\theta} &= (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}, && (51) \end{aligned}$$

which is the equation for solving θ. This is called the Normal Equation for linear regression.